CPathoGen: A Framework for Controllable Histopathology Generation for Probing Pathology Foundation Models

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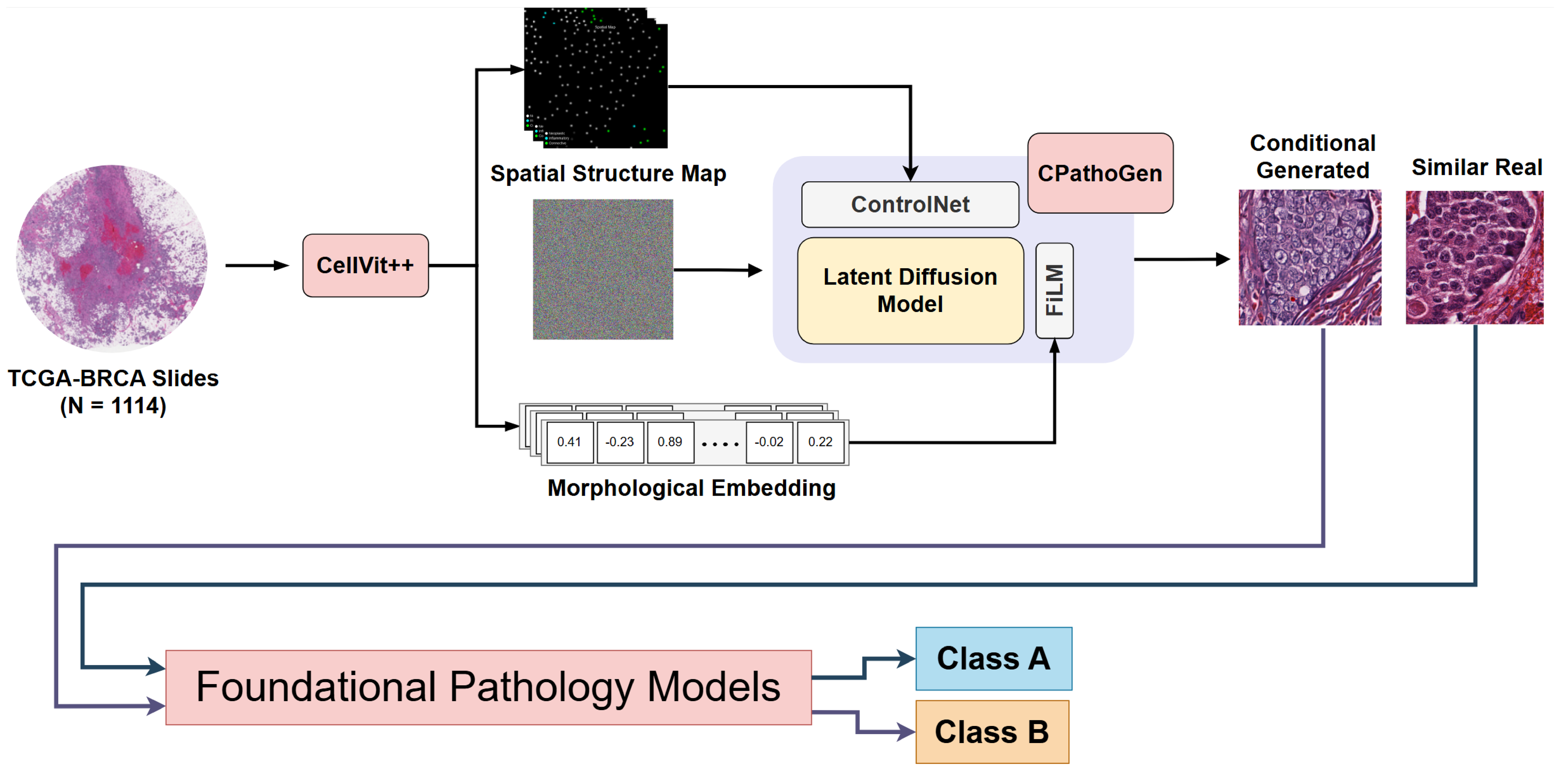
Introduction

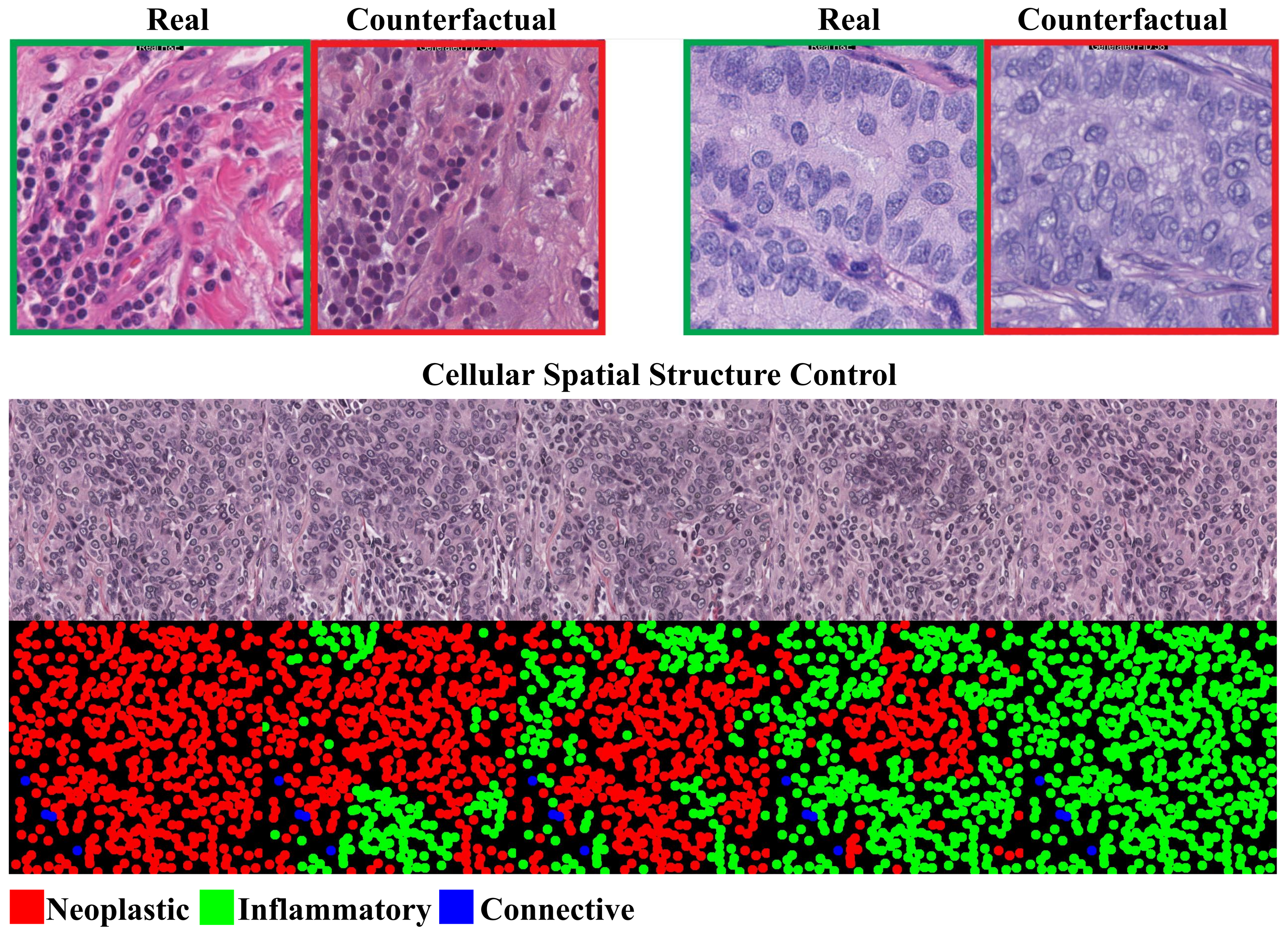
Problem: Foundation models show strong performance in pathology, but accuracy alone does not ensure they rely on biologically meaningful features. Models may instead depend on stain color, artifacts, or other shortcuts, creating a key trust and explainability challenge that current evaluation methods rarely test directly.

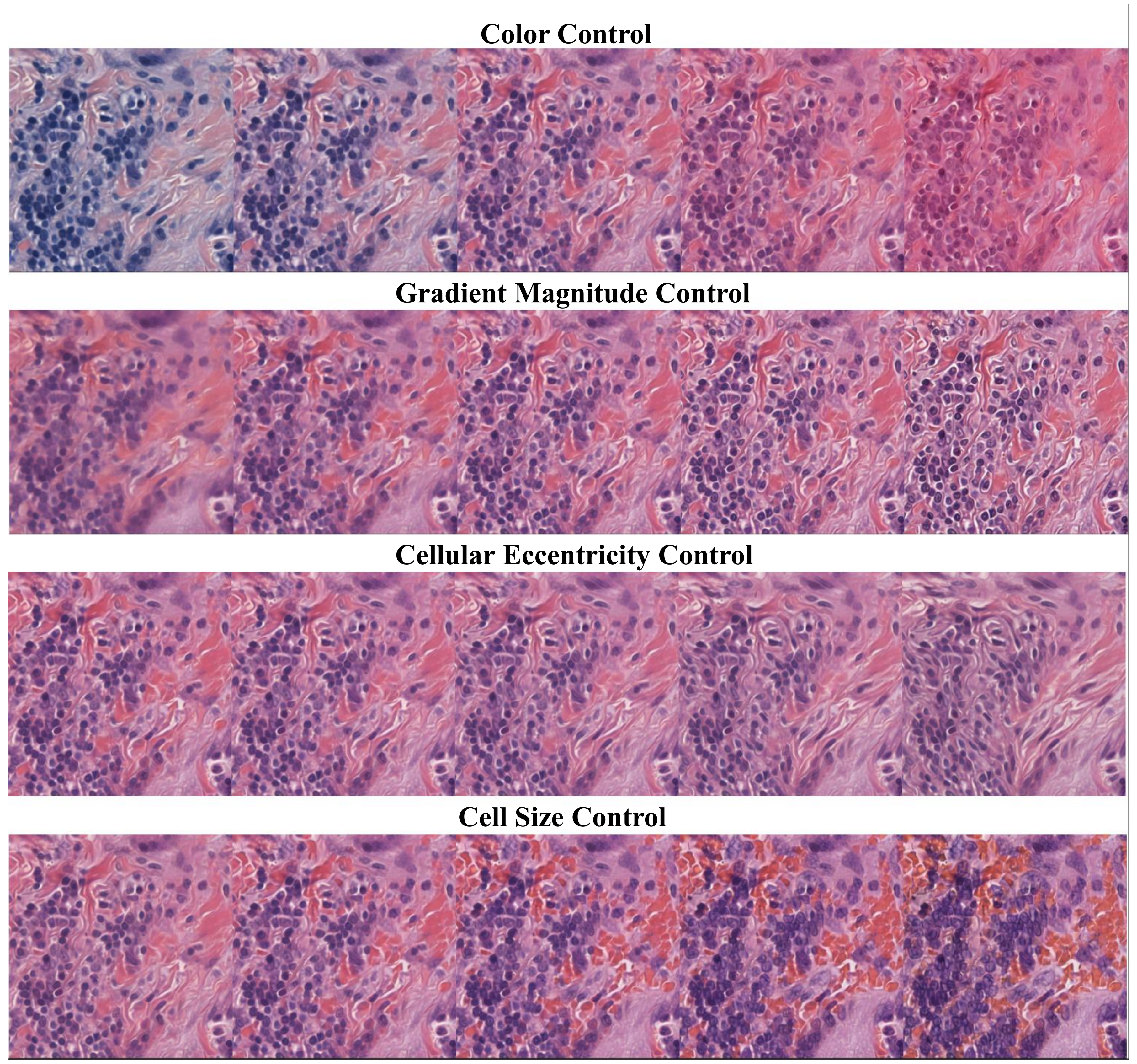
Methodology: We developed CPathoGen, a controllable latent diffusion framework for Hematoxylin and Eosin (H&E) stained tiles generation. Weak cell annotations were used to create cell-type spatial layout maps and to extract tumor morphology features. A pretrained latent diffusion backbone was adapted to H&E by training on 1 million tiles, with ControlNet used for spatial layout conditioning and Feature-wise Linear Modulation used for morphology conditioning (FID ~56). We then used generated counterfactuals to probe top pathology foundational models on breast cancer subtyping tasks.

Pipeline Overview



Results





Experiments

Table 1: Molecular Subtyping Benchmark Results (Basal, HER2, Luminal)

	UNI-2h				CTransPath				ResNet50			
Metric	**LR**	**RF**	**XGB**	**MLP**	**LR**	**RF**	**XGB**	**MLP**	**LR**	**RF**	**XGB**	**MLP**
Accuracy (Tile)	0.581	0.537	0.562	0.584	0.540	0.537	0.535	0.530	0.417	0.402	0.392	0.414
Accuracy (Pat-Mean)	0.699	0.639	0.654	0.697	0.626	0.629	0.616	0.606	0.450	0.447	0.404	0.475
Accuracy (Pat-Max)	0.697	0.634	0.651	0.689	0.626	0.619	0.603	0.608	0.427	0.442	0.437	0.447
AUC (Macro)	0.755	0.724	0.751	0.764	0.720	0.720	0.716	0.713	0.604	0.586	0.578	0.596

Table 2: Model Sensitivity to Synthetic Morphological and Color Perturbations

	Invariance Rate		Prediction Flip Rate	
Foundation Model	**Color**	**Morphology**	**Color**	**Morphology**
UNI-2h	0.25	0.49	0.75	0.51
CTransPath	0.34	0.61	0.66	0.39
ResNet50	0.58	0.68	0.42	0.32

Conclusion

Conclusion: CPathoGen enables the generation of realistic breast cancer H&E images with controllable cellular layouts and tumor morphology, providing a framework for systematic counterfactual testing. Using these generated perturbations, we found that UNI-2h achieved the strongest subtype classification performance but showed substantial sensitivity to color-based counterfactual changes, suggesting that high predictive accuracy may coexist with vulnerability to appearance shifts in generated H&E images.